

Task 3

(1)

(1.1)

- Jaccard 1->2: 0.3036496
- Jaccard 2-3: 0.3506686

(1.2)

```
# 1. Create the initial effects object based on your data
myeff <- getEffects(mydata)

# 2. Add Endogenous Controls (Network Structure)
# Note: outdegree (density) and recip (reciprocity) are already included by default
myeff <- includeEffects(myeff, transTrip, cycle3)

# 3. Add Exogenous Controls (Demographics)
# Controlling for gender dynamics
myeff <- includeEffects(myeff, egoX, altX, sameX, interaction1 = "gender")

# Controlling for base alcohol tendencies (sender and receiver effects)
myeff <- includeEffects(myeff, egoX, altX, interaction1 = "alcohol_covar")
# (Optional) You can also add age controls if desired:
# myeff <- includeEffects(myeff, simX, interaction1 = "age")

# 4. Add Hypothesis Tests!
# H(i): Popularity
myeff <- includeEffects(myeff, inPop)

# H(ii): Alcohol Similarity
myeff <- includeEffects(myeff, simX, interaction1 = "alcohol_covar")

# H(iii): Distance / Proximity
myeff <- includeEffects(myeff, X, interaction1 = "distance")
```

(1.3)

##	dependent	effect	theta	s.e.	p.value	sig.
## 1	rate constant friendship rate (period 1)		11.014	1.037		
## 2	rate constant friendship rate (period 2)		9.006	0.811		
## 3	friendship	outdegree (density)	-3.074	0.128	0	***
## 4	friendship	reciprocity	2.207	0.109	0	***

```
## 5 friendship      transitive triplets  0.687 0.043      0 ***
## 6 friendship      3-cycles -0.467 0.087      0 ***
## 7 friendship      indegree - popularity -0.080 0.020      0 ***
## 8 friendship      distance -0.160 0.046      0.001 ***
## 9 friendship      gender alter -0.177 0.104      0.089 .
## 10 friendship     gender ego  0.145 0.107      0.176
## 11 friendship     same gender  0.899 0.097      0 ***
## 12 friendship     alcohol_covar alter  0.032 0.032      0.318
## 13 friendship     alcohol_covar ego -0.024 0.031      0.443
## 14 friendship     alcohol_covar similarity  0.647 0.167      0 ***
##      t.conv
## 1
## 2
## 3      0.01
## 4     -0.019
## 5     -0.051
## 6     -0.052
## 7     -0.003
## 8      0.005
## 9     -0.044
## 10    -0.029
## 11     0.018
## 12     0.006
## 13     0.021
## 14     0.033
```

```
t.conv <- apply(model0$sf, 2, mean) / apply(model0$sf, 2, sd)
t.conv
```

```
## [1] 0.010011518 -0.019461490 -0.050976298 -0.051510397 -0.003137504
## [6] 0.005368174 -0.044060343 -0.028901295 0.017515469 0.006034284
## [11] 0.021009166 0.032584853
```

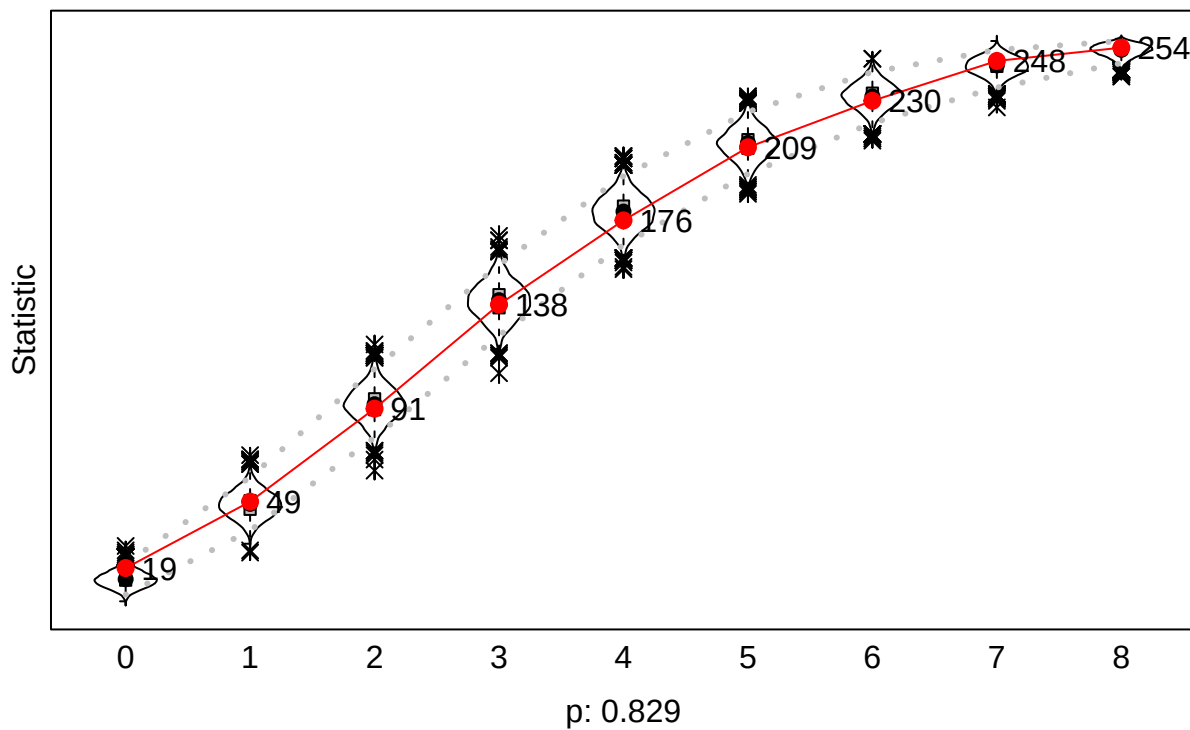
```
overall <- sqrt(t(apply(model0$sf, 2, mean)) %*% solve(cov(model0$sf)) %*% apply(model0$sf, 2, mean))
overall
```

```
##      [,1]
## [1,] 0.1204442
```

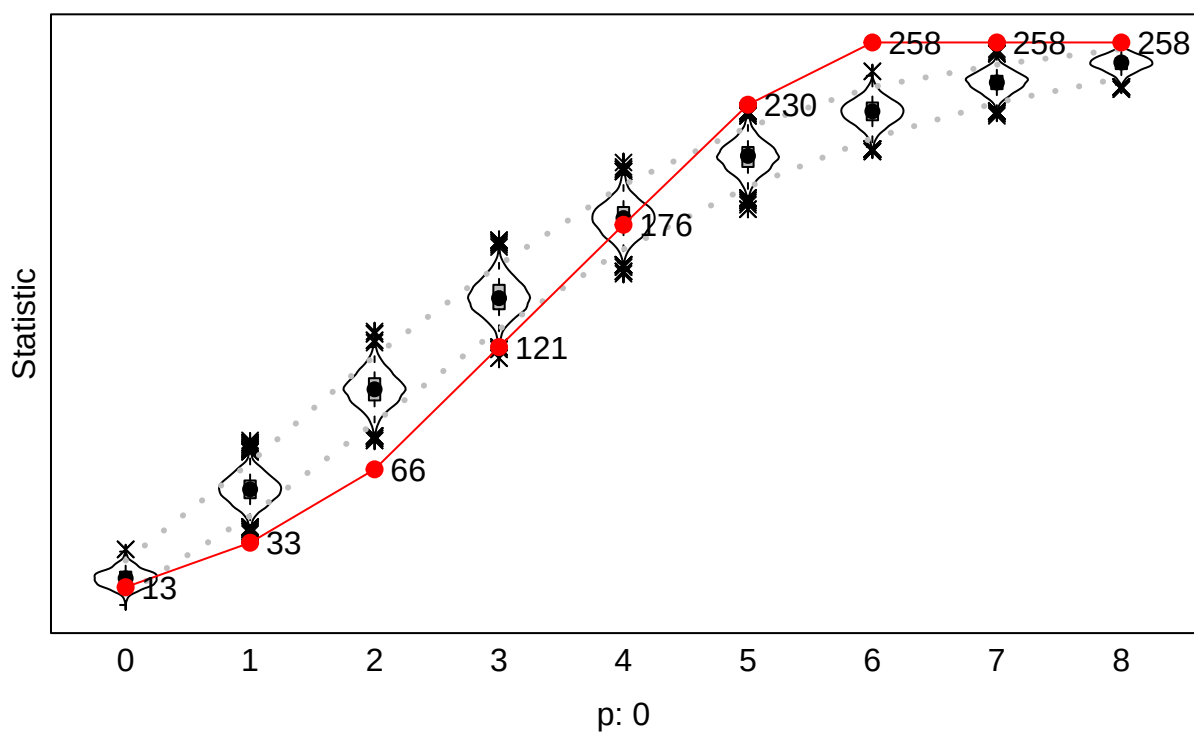
We can see that the model converges.

(1.4)

Goodness of Fit of IndegreeDistribution

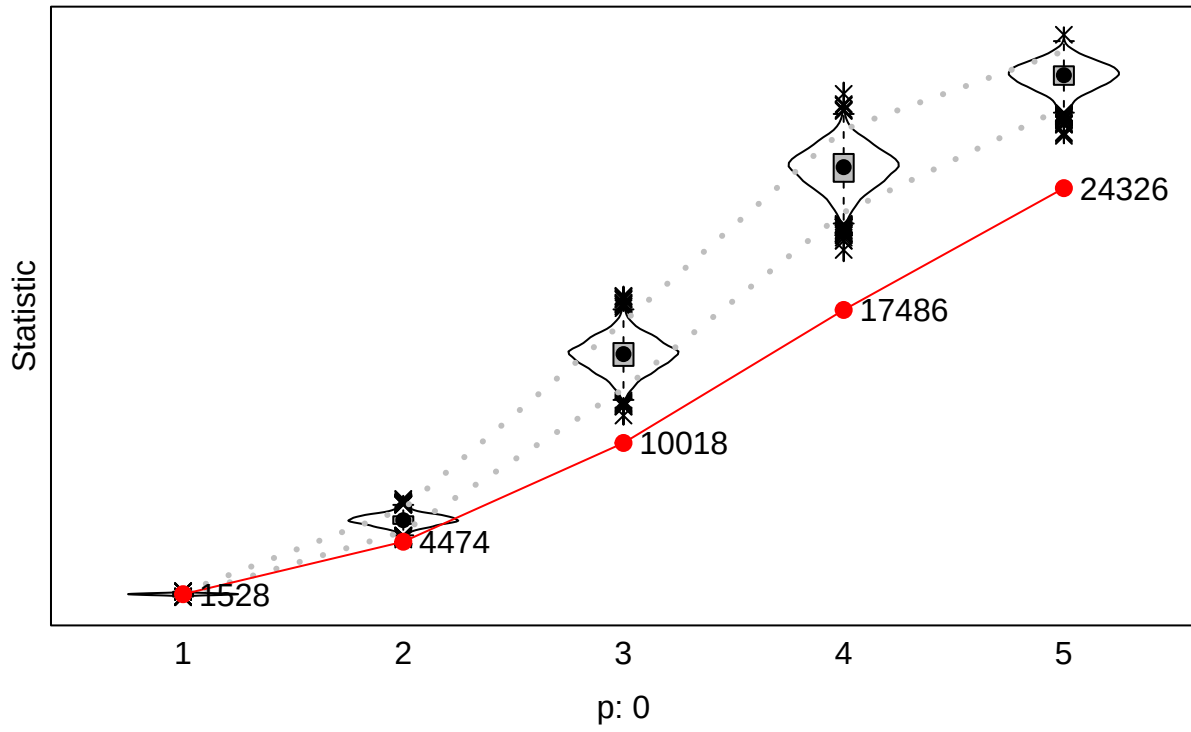


Goodness of Fit of OutdegreeDistribution

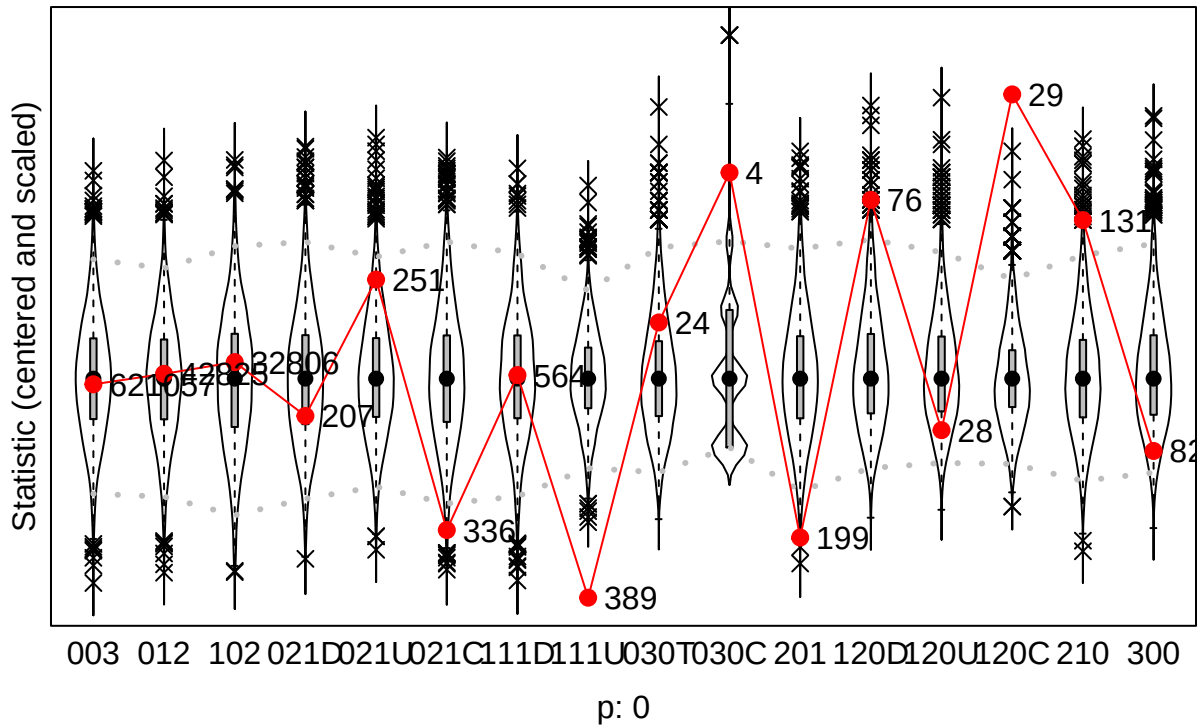


```
## Note: some statistics are not plotted because their variance is 0.  
## This holds for the statistic: Inf.
```

Goodness of Fit of GeodesicDistribution



Goodness of Fit of TriadCensus



We can see that the model has good GOF in terms of in-degree distribution, but everything else is quite poor. This means that the model does not represent the data well.

From now on (for interpretation purposes), we pretend that the model has good GOF, so we can draw conclusions.

(1.5)

- i - The coefficient is negative and significant. We reject the hypothesis, since we observe the opposite effect.
- ii - positive and significant - hypothesis is supported
- iii - the coefficient is negative and significant - hypothesis is supported

(2)

(2.1)

```
# =====
# SELECTION PART (Network Evolution)
# =====
# Endogenous Controls
myeff <- set_effect(myeff, list(transTrip, cycle3), depvar="friendship")
```

```

# Exogenous Controls (Gender)
myeff <- set_effect(myeff, list(egoX, altX, sameX), covar1="gender", depvar="friendship")

# Hypothesis (i): Popularity -> Friendship
myeff <- set_effect(myeff, list(inPop), depvar="friendship")

# Hypothesis (ii): Alcohol Similarity -> Friendship (Homophily)
# Note: covar1 is now targeting our new behavior variable "alcohol_beh"
myeff <- set_effect(myeff, list(simX, egoX, altX), covar1="alcohol_beh", depvar="friendship")

# Hypothesis (iii): Distance -> Friendship
myeff <- set_effect(myeff, list(X), covar1="distance", depvar="friendship")

# =====
# INFLUENCE PART (Behavior Evolution)
# =====
# Note: RSiena automatically includes the 'linear' and 'quad' (shape)
# effects for behavior, which act as your basic endogenous controls.

# Exogenous Control: Gender's effect on alcohol consumption
# (It's good practice to control for basic demographics in behavior too)
myeff <- set_effect(myeff, list(effFrom), covar1="gender", depvar="alcohol_beh")

# Hypothesis (iv): Popularity -> Alcohol
# Does indegree in the friendship network predict alcohol behavior?
myeff <- set_effect(myeff, list(indeg), covar1="friendship", depvar="alcohol_beh")

# Hypothesis (v): Assimilation / Social Influence
# Do students adjust alcohol consumption to be similar to their friends?
myeff <- set_effect(myeff, list(avSim), covar1="friendship", depvar="alcohol_beh")

```

(2.2)

```

t.conv <- apply(model0$sf, 2, mean) / apply(model0$sf, 2, sd)
t.conv

## [1] 0.010011518 -0.019461490 -0.050976298 -0.051510397 -0.003137504
## [6] 0.005368174 -0.044060343 -0.028901295 0.017515469 0.006034284
## [11] 0.021009166 0.032584853

overall <- sqrt(t(apply(model0$sf, 2, mean)) %*% solve(cov(model0$sf)) %*% apply(model0$sf, 2, mean))
overall

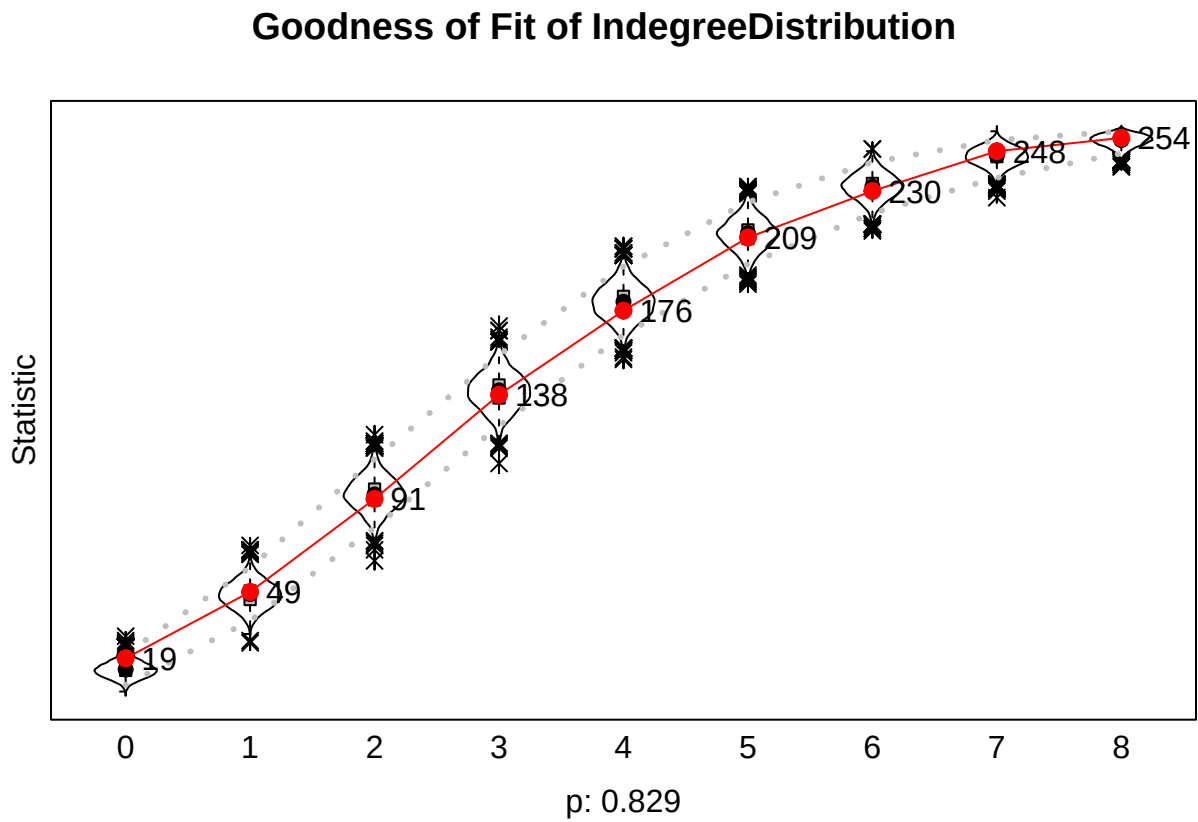
## [1,]
## [1,] 0.1204442

```

The model converges.

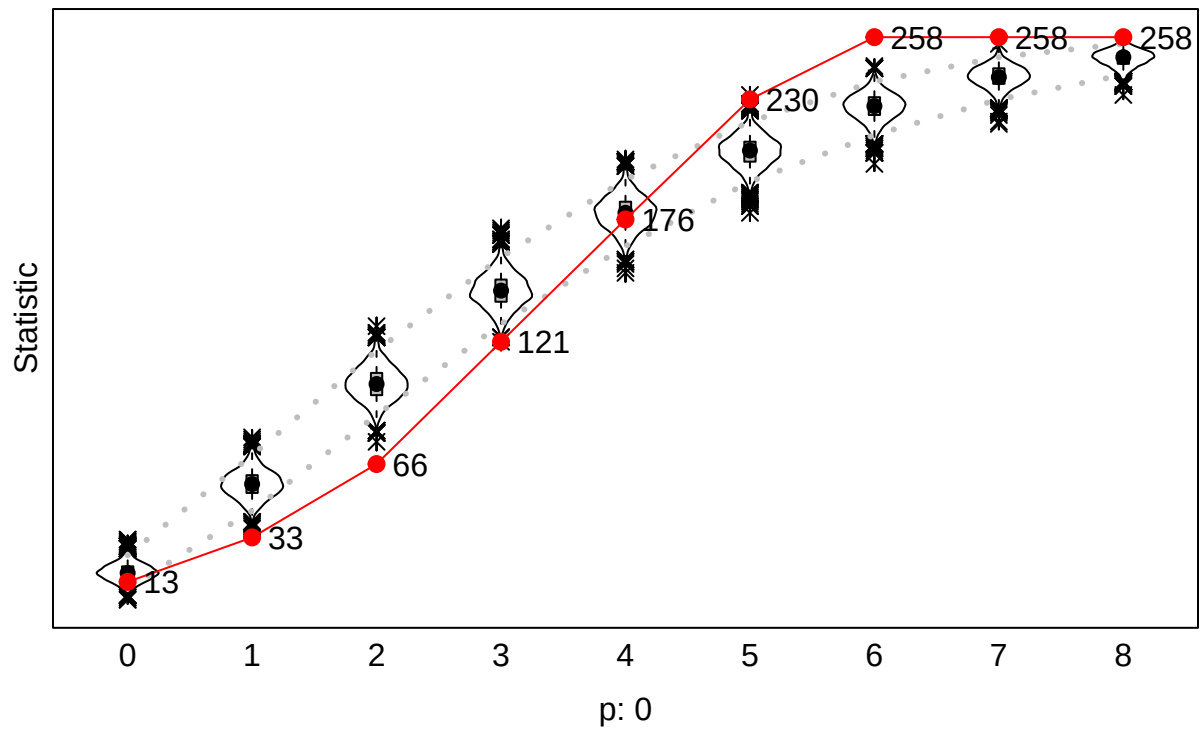
(2.3)

```
gofCoevId <- readRDS("task3_modelCoev_gofCoevId.rds")  
plot(gofCoevId)
```



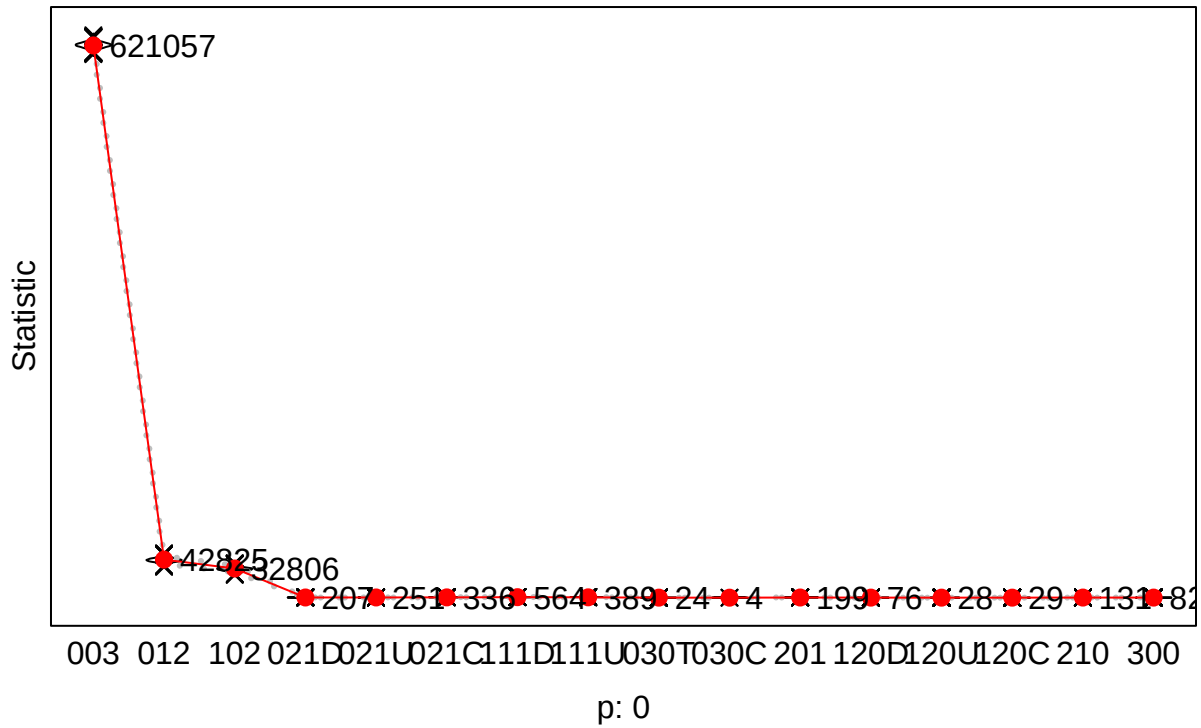
```
# Outdegree distribution  
gofCoevOd <- readRDS("task3_modelCoev_gofCoevOd.rds")  
plot(gofCoevOd)
```

Goodness of Fit of OutdegreeDistribution



```
# Triad census
gofCoevTC <- readRDS("task3_modelCoev_gofCoevTC.rds")
plot(gofCoevTC)
```

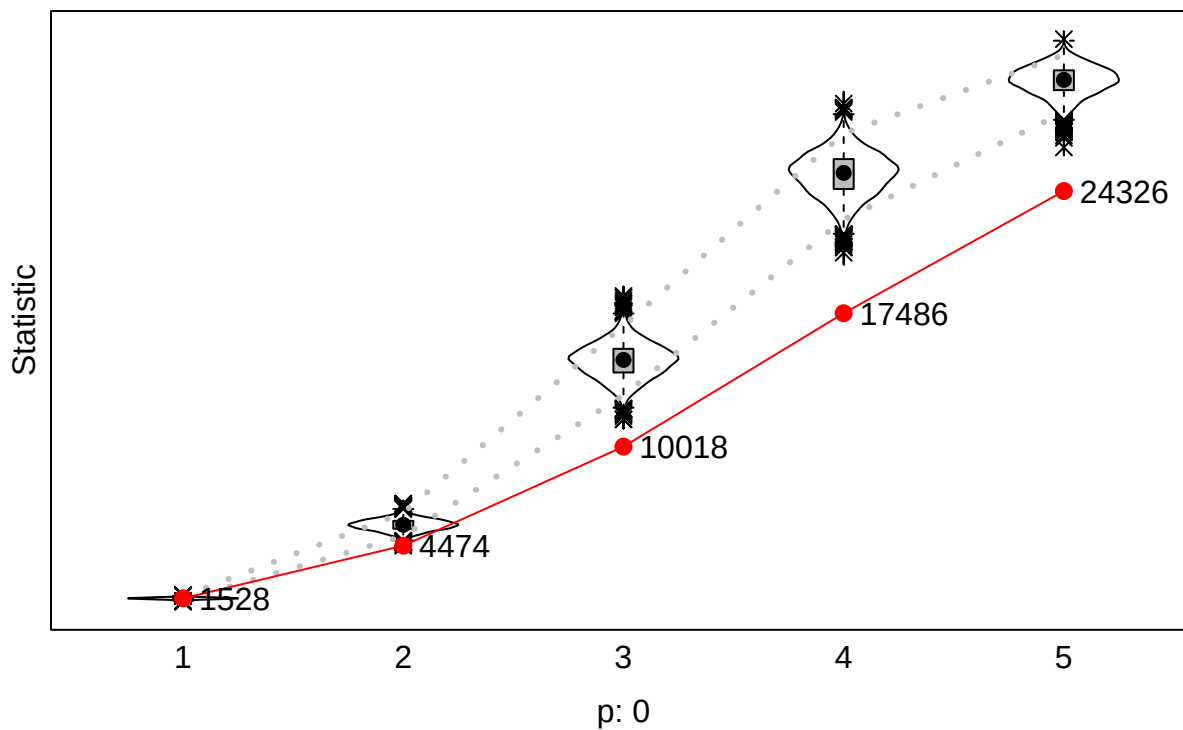

Goodness of Fit of TriadCensus



```
# Geodesic distance
gofCoevGD <- readRDS("task3_modelCoev_gofCoevGD.rds")
plot(gofCoevGD)
```

```
## Note: some statistics are not plotted because their variance is 0.
## This holds for the statistic: Inf.
```

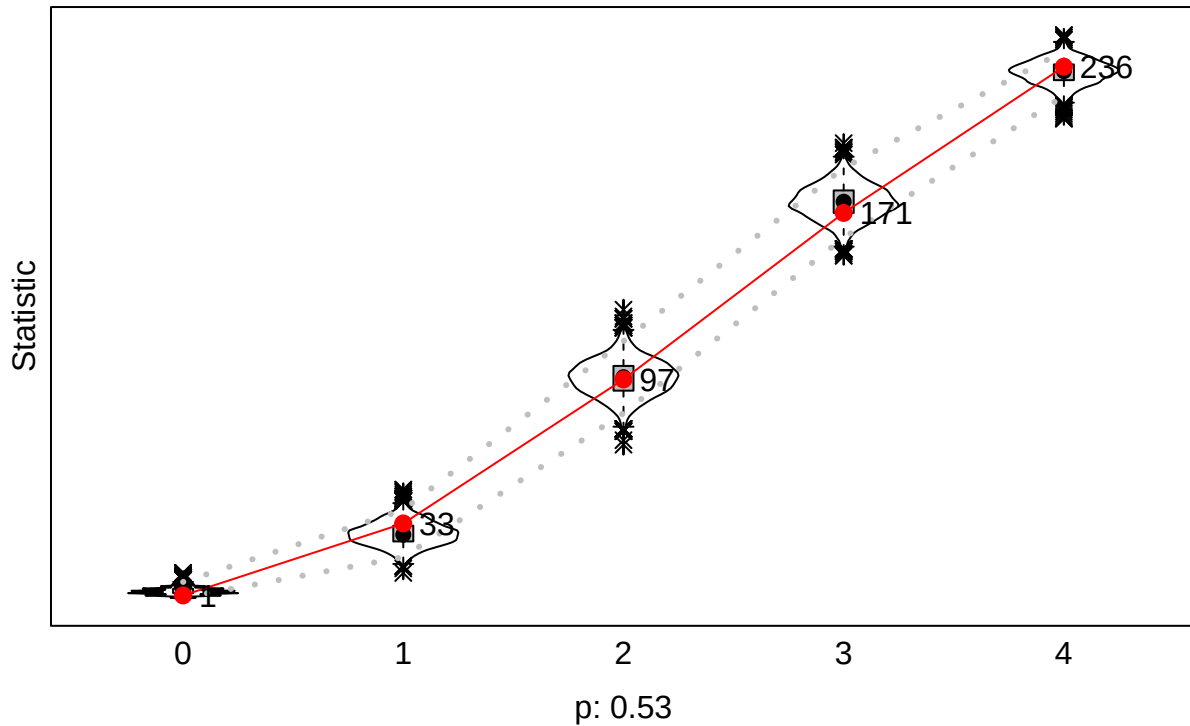
Goodness of Fit of GeodesicDistribution



```
# Behaviour distribution
gofCoevBeh <- readRDS("task3_modelCoev_gofCoevBeh.rds")
plot(gofCoevBeh)
```

```
## Note: some statistics are not plotted because their variance is 0.
## This holds for the statistic: 5.
```

Goodness of Fit of BehaviorDistribution



GOF of in-degree distribution is ok, the rest is not good. The model does not represent the data very well. We (again) assume that the model does assume the data, so we can interpret what we have

(2.4)

```
printSienaCoev(modelCoev)
```

##		effect	theta	s.e.	p.value	sig.	t.conv
## 1	constant friendship rate (period 1)	10.941	1.049				
## 2	constant friendship rate (period 2)	8.922	0.838				
## 3	outdegree (density)	-3.092	0.131	0	***	-0.033	
## 4	reciprocity	2.178	0.107	0	***	-0.01	
## 5	transitive triplets	0.683	0.043	0	***	-0.017	
## 6	3-cycles	-0.459	0.085	0	***	-0.015	
## 7	indegree - popularity	-0.081	0.020	0	***	-0.031	
## 8	distance	-0.164	0.046	0	***	0.021	
## 9	gender alter	-0.174	0.107	0.104		0.016	
## 10	gender ego	0.147	0.107	0.168		0.026	
## 11	same gender	0.895	0.098	0	***	-0.039	
## 12	alcohol_beh alter	0.047	0.049	0.334		0.004	
## 13	alcohol_beh ego	-0.046	0.048	0.337		0.004	
## 14	alcohol_beh similarity	1.195	0.354	0.001	***	-0.013	
## 15	rate alcohol_beh (period 1)	1.646	0.271				

## 16	rate alcohol_beh (period 2)	2.355	0.395			
## 17	alcohol_beh linear shape	0.137	0.245	0.576		-0.039
## 18	alcohol_beh quadratic shape	0.036	0.070	0.608		-0.029
## 19	alcohol_beh average similarity	6.960	2.024	0.001	***	-0.022
## 20	alcohol_beh indegree	0.071	0.065	0.276		-0.031
## 21	alcohol_beh: effect from gender	-0.023	0.185	0.901		0.012

- iv - coefficient is insignificant. cannot confirm hypothesis.
- v - coefficient is positive and significant - confirm hypothesis

(2.5)

We see both selection and influence effects: - Pupils with similar alcohol consumption become friends, but also friends change their alcohol consumption to be similar

We also notice that these selection processes are also in place: - Students become friends with other popular students - Reciprocity

(3)

- We can add `outActSqrt`. It is closely related to the out-degree distribution and should help us steer the model in the right direction
- We can add `gvespFF` to control for the amount of shared partners. This will help us improve the distance distribution.

(4)

- Students with many friends tend to increase or maintain alcohol consumption
- Students tend to adjust their literacy score results towards the ones of their friends